FEATURES

Features are engineered variables that extract meaningful patterns from raw sensor data. A temperature reading of 185°C becomes meaningful when transformed into features like "temperature deviation from optimal", "rate of temperature change", or "time since temperature exceeded safe operating range". These features capture relationships and contextual information that individual measurements cannot express.

Figure 4.3 shows the relationship between raw data stores and features as used both by a data scientist to train an ML model, and also for ML model serving.

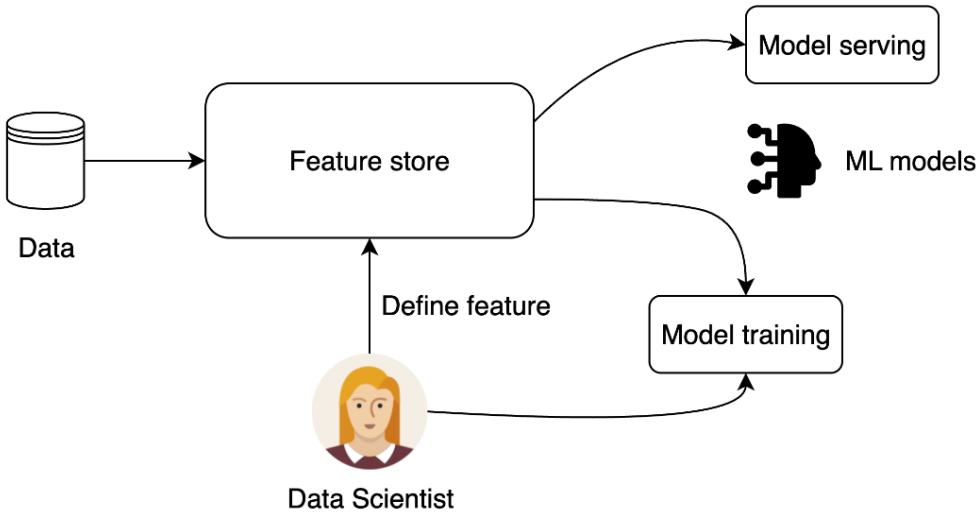


Figure 4.3 The relationship between a raw data store, derived data in a feature store, and the training and serving of ML models.

EMBEDDINGS

Embeddings are dense, numerical vector representations that encode complex patterns, relationships, and semantics from raw data into a compact, mathematically comparable format. Unlike traditional features that extract specific measurable attributes, embeddings capture semantic relationships within high-dimensional data. Figure 4.4 shows a simplified example of how raw data might be represented as a 2D vector, illustrating how sensor readings might be converted into a format where a mathematical operation (distance between two points) corresponds to a real world relationship.